D51 | Shares, then elections

Fictional INR-crore financing case | Share units are arbitrary | 13 November 2026 | Read the original D51 case before using this worked check.

Rebuild before checking	Use a blank spreadsheet to reconstruct each round. The SAFE-stage percentages are implied conversion economics, not issued voting stock. Economic dilution does not establish board or voting control.
Fixed classroom assumptions	Founders 100; post-money SAFE 2 at 20 and 3 at 30; both caps bind; no interest, discount, MFN, side letter or pro rata. A 25 at 100 FD pre-money; new pool 12% of post-A is funded within pre-money. B 80 at 320 FD pre-money, no top-up or new instruments.
Pool treatment at exit	The entire pool remains unissued and is cancelled without consideration. It receives zero and is excluded from conversion denominators. No employee has an issued or promised claim in this fictional case.
One class, one election	Each investor class chooses optional 1x non-participating preference OR conversion, never both. B is senior to A; A senior to equal-ranking seed classes. If seed preference cash is short, divide available cash pro rata to the two elected preference claims.
Test all elections	Each exit sheet lists all 16 profiles. A flag of 1 means preference and 0 conversion. A stable row gives no investor a larger payout by changing only its own election. Compare the alternative row, not its headline 1x amount alone.
Sensitivity and limits	Change only the sensitivity net exit value on Inputs. This changes an economic assumption, not real contracts. If multiple profiles are stable, inspect each; ties are not a unique recommendation. No account, fundraising, securities issuance, document sending or submission is needed.
Authored execution only	Native recalculation checks show this authored model works under these assumptions. They do not establish Excel/Sheets parity, real security enforceability, Indian requirements, actual fundraising outcomes or independent learner transfer.

Worked source basis: YC post-money SAFE user guide, quick start and Q&A B/C. Optional preference rules and seniority here are stipulated classroom terms, not claimed universal document terms.

Inputs | Keep the denominator explicit

Only the sensitivity exit value is yellow/editable. Other displayed numbers are the fixed D51 case. Net exits are after all debt, tax and transaction costs.

Founders initial shares	100.0000
Seed1 investment	2.0000
Seed1 post-money cap	20.0000
Seed2 investment	3.0000
Seed2 post-money cap	30.0000
A investment	25.0000
A FD pre-money valuation	100.0000
New pool / post-A FD	12.00%
B investment	80.0000
B FD pre-money valuation	320.0000
High net equity exit	600.0000
Low net equity exit	150.0000
Sensitivity net equity exit	300.00

Use non-negative net equity proceeds. At zero, every profile ties at zero and is weakly stable. Stability is checked with a tiny floating-point tolerance of 1e-8 crore; economically exact ties are not strict advantages.

Shares | Reconcile counts and ownership

FD includes unissued pool in financing rounds. After cancelling the unissued pool, only actual converted shares participate at exit. Fixed cap-binding case, not a generalized SAFE engine.

Holder	SAFE implied count	SAFE implied %	Post-A count	Post-A FD %	Post-B count	Post-B FD %	Exit converted %
Founders	100.0000	80.00%	100.0000	54.40%	100.0000	43.52%	48.14%
Seed1	12.5000	10.00%	12.5000	6.80%	12.5000	5.44%	6.02%
Seed2	12.5000	10.00%	12.5000	6.80%	12.5000	5.44%	6.02%
A	0.0000	0.00%	36.7647	20.00%	36.7647	16.00%	17.70%
B	0.0000	0.00%	0.0000	0.00%	45.9559	20.00%	22.12%
Unissued pool	0.0000	0.00%	22.0588	12.00%	22.0588	9.60%	0.00%
Totals	125.0000	100.00%	183.8235	100.00%	229.7794	100.00%	100.00%

A price per share	0.6800
Seed1 conversion price	0.1600
Seed2 conversion price	0.2400
Cap binding price check	1.0000
B price per share	1.7408

A price includes conversion shares and the new pre-money-funded pool. Compare actual conversion prices, not cap/pre-money ratios alone. Founders fall below 50% at B, not below 25% in these rounds. Shares are arbitrary units; cash amounts are crore.

Exit600 | Test all sixteen elections

1 = take preference; 0 = convert. Flags are ordered B, A, Seed1, Seed2. Each row is a complete exclusive-election outcome; stable means no profitable unilateral change.

Profile	B pref?	A pref?	S1 pref?	S2 pref?	Founders	Seed1	Seed2	A payout	B payout	Total	Stable?
0	0	0	0	0	288.8496	36.1062	36.1062	106.1947	132.7434	600.0000	1.0000
1	0	0	0	1	305.8079	38.2260	3.0000	112.4294	140.5367	600.0000	0.0000
2	0	0	1	0	306.3202	2.0000	38.2900	112.6177	140.7721	600.0000	0.0000
3	0	0	1	1	325.6338	2.0000	3.0000	119.7183	149.6479	600.0000	0.0000
4	0	1	0	0	336.3441	42.0430	42.0430	25.0000	154.5699	600.0000	0.0000
5	0	1	0	1	360.9838	45.1230	3.0000	25.0000	165.8933	600.0000	0.0000
6	0	1	1	0	361.6148	2.0000	45.2019	25.0000	166.1833	600.0000	0.0000
7	0	1	1	1	390.5290	2.0000	3.0000	25.0000	179.4710	600.0000	0.0000
8	1	0	0	0	321.4545	40.1818	40.1818	118.1818	80.0000	600.0000	0.0000
9	1	0	0	1	346.3645	43.2956	3.0000	127.3399	80.0000	600.0000	0.0000
10	1	0	1	0	347.0345	2.0000	43.3793	127.5862	80.0000	600.0000	0.0000
11	1	0	1	1	376.5591	2.0000	3.0000	138.4409	80.0000	600.0000	0.0000
12	1	1	0	0	396.0000	49.5000	49.5000	25.0000	80.0000	600.0000	0.0000
13	1	1	0	1	437.3333	54.6667	3.0000	25.0000	80.0000	600.0000	0.0000
14	1	1	1	0	438.2222	2.0000	54.7778	25.0000	80.0000	600.0000	0.0000
15	1	1	1	1	490.0000	2.0000	3.0000	25.0000	80.0000	600.0000	0.0000

Stable profiles (ties allowed)	1.0000

Read every green row. No 1x preference is added to a converted payout. Other holders' elections stay fixed in each unilateral comparison. Hidden N:S columns expose waterfall helpers when unhidden; no separate election optimization add-in is required.

Exit150 | Test all sixteen elections

1 = take preference; 0 = convert. Flags are ordered B, A, Seed1, Seed2. Each row is a complete exclusive-election outcome; stable means no profitable unilateral change.

Profile	B pref?	A pref?	S1 pref?	S2 pref?	Founders	Seed1	Seed2	A payout	B payout	Total	Stable?
0	0	0	0	0	72.2124	9.0265	9.0265	26.5487	33.1858	150.0000	0.0000
1	0	0	0	1	75.2994	9.4124	3.0000	27.6836	34.6045	150.0000	0.0000
2	0	0	1	0	75.8117	2.0000	9.4765	27.8719	34.8399	150.0000	0.0000
3	0	0	1	1	79.3561	2.0000	3.0000	29.1751	36.4688	150.0000	0.0000
4	0	1	0	0	73.1183	9.1398	9.1398	25.0000	33.6022	150.0000	0.0000
5	0	1	0	1	76.9930	9.6241	3.0000	25.0000	35.3828	150.0000	0.0000
6	0	1	1	0	77.6241	2.0000	9.7030	25.0000	35.6729	150.0000	0.0000
7	0	1	1	1	82.2166	2.0000	3.0000	25.0000	37.7834	150.0000	0.0000
8	1	0	0	0	43.2727	5.4091	5.4091	15.9091	80.0000	150.0000	0.0000
9	1	0	0	1	44.8867	5.6108	3.0000	16.5025	80.0000	150.0000	0.0000
10	1	0	1	0	45.5567	2.0000	5.6946	16.7488	80.0000	150.0000	0.0000
11	1	0	1	1	47.5269	2.0000	3.0000	17.4731	80.0000	150.0000	0.0000
12	1	1	0	0	36.0000	4.5000	4.5000	25.0000	80.0000	150.0000	1.0000
13	1	1	0	1	37.3333	4.6667	3.0000	25.0000	80.0000	150.0000	0.0000
14	1	1	1	0	38.2222	2.0000	4.7778	25.0000	80.0000	150.0000	0.0000
15	1	1	1	1	40.0000	2.0000	3.0000	25.0000	80.0000	150.0000	0.0000

Stable profiles (ties allowed)	1.0000

Read every green row. No 1x preference is added to a converted payout. Other holders' elections stay fixed in each unilateral comparison. Hidden N:S columns expose waterfall helpers when unhidden; no separate election optimization add-in is required.

Sensitivity | Test all sixteen elections

1 = take preference; 0 = convert. Flags are ordered B, A, Seed1, Seed2. Each row is a complete exclusive-election outcome; stable means no profitable unilateral change.

Profile	B pref?	A pref?	S1 pref?	S2 pref?	Founders	Seed1	Seed2	A payout	B payout	Total	Stable?
0	0	0	0	0	144.4248	18.0531	18.0531	53.0973	66.3717	300.0000	0.0000
1	0	0	0	1	152.1356	19.0169	3.0000	55.9322	69.9153	300.0000	0.0000
2	0	0	1	0	152.6478	2.0000	19.0810	56.1205	70.1507	300.0000	0.0000
3	0	0	1	1	161.4487	2.0000	3.0000	59.3561	74.1952	300.0000	0.0000
4	0	1	0	0	160.8602	20.1075	20.1075	25.0000	73.9247	300.0000	0.0000
5	0	1	0	1	171.6566	21.4571	3.0000	25.0000	78.8863	300.0000	0.0000
6	0	1	1	0	172.2877	2.0000	21.5360	25.0000	79.1763	300.0000	0.0000
7	0	1	1	1	184.9874	2.0000	3.0000	25.0000	85.0126	300.0000	0.0000
8	1	0	0	0	136.0000	17.0000	17.0000	50.0000	80.0000	300.0000	1.0000
9	1	0	0	1	145.3793	18.1724	3.0000	53.4483	80.0000	300.0000	0.0000
10	1	0	1	0	146.0493	2.0000	18.2562	53.6946	80.0000	300.0000	0.0000
11	1	0	1	1	157.2043	2.0000	3.0000	57.7957	80.0000	300.0000	0.0000
12	1	1	0	0	156.0000	19.5000	19.5000	25.0000	80.0000	300.0000	0.0000
13	1	1	0	1	170.6667	21.3333	3.0000	25.0000	80.0000	300.0000	0.0000
14	1	1	1	0	171.5556	2.0000	21.4444	25.0000	80.0000	300.0000	0.0000
15	1	1	1	1	190.0000	2.0000	3.0000	25.0000	80.0000	300.0000	0.0000

Stable profiles (ties allowed)	1.0000

Read every green row. No 1x preference is added to a converted payout. Other holders' elections stay fixed in each unilateral comparison. Hidden N:S columns expose waterfall helpers when unhidden; no separate election optimization add-in is required.

Practice | Explain the mechanism

Use your own blank model before checking these worked targets. Yellow answer spaces stay empty; no automatic grading, tracking or submission.

Baseline check	Your result	Target	Difference
Founders post-A FD		0.5440	
Founders post-B FD		0.4352	
Pool post-B FD		0.0960	
Low exit founders		36.0000	
Low exit Seed1		4.5000	
Low exit Seed2		4.5000	
Low exit A		25.0000	
Low exit B		80.0000	

At 600, profile 0 converts all classes. At 150, profile 12 takes B/A preferences and converts both seeds. Check unilateral deviations, not just whether conversion exceeds original investment in the all-converted row.

At the initial 300 sensitivity, find the stable profile before looking at the highlights. Then try 100, 20, 3 and 0 to test seniority, exhausted proceeds, pari-passu seed allocation and ties. Restore 300 before saving your baseline.

Source: https://www.ycombinator.com/assets/ycdc/Primer%20for%20post-money%20safe%20v1.1-2af8129e12effd9638eeab383b7309142c8f415e5cdb0bc210d573f779177a1c.pdf | U.S. educational mechanics only. This case stipulates election rights/seniority; real agreements and current local law govern real financings.